

Here is the assignment instruction:

Assignment: Using a New York Times API in R

The New York Times provides multiple public APIs via its developer portal (NYT Developer Network). To use any NYT API, you must first create an account and obtain an API key.

Your task:

Choose one New York Times API endpoint (e.g., Article Search, Most Popular, Books, etc.). You should use your data analysis to ask and attempt to answer an interesting question of your choosing. Examples: "What are the top five best selling hard cover books?", "Which newspaper sections have produced the most popular articles, and how has this changed compared to X years ago?", etc.

In R, write code to:

Authenticate using your API key. Never hard code an API key. There are a number of good ways to do this. For example: Store it in an environment variable (e.g., `Sys.getenv("NYT_API_KEY")`).

Make a request to the endpoint

Parse the JSON response

Transform the result into a clean R data frame (tibble is fine)

Deliverable

A single Quarto (.qmd) document that includes:

The API you selected (endpoint + brief description)

The request you made (parameters used)

Code that returns a tidy data frame

A short note describing any data-cleaning decisions (e.g., nested fields, missing values)

I completed the following part in the approach submission:

title: "Assignment NY Times API"

author: "Khandker Qaiduzzaman"

format: html

editor: visual

Objective

The objective of this assignment is to use a New York Times API to retrieve JSON data, import it into R, and transform it into a tidy data frame for analysis. The goal is to explore patterns in article engagement by examining whether articles with images appear more frequently in the New York Times Most Popular list compared to those without images.

Approach

For this assignment, I used the New York Times Most Popular API, specifically the endpoint that returns the most viewed articles over the past 7 days.

The research question for this analysis is:

- ****Do articles with images receive more engagement than those without in the New York Times Most Popular list?****

The dataset should be obtained dynamically through an API request and include information about the most viewed articles published by The New York Times. To answer the research question, the media field will be used to determine whether an article contains images. A new variable will be created:

****Has Image –**** A binary variable indicating whether the article includes at least one image (Yes/No)

The data retrieved from the API is in JSON format, which represents hierarchical and nested data commonly used in web services. This requires parsing and transformation before it can be analyzed in R.

Data Analysis Steps

The analysis will follow these steps:

1. ****API Authentication:**** The API key will be securely

accessed using an environment variable (Sys.getenv) to avoid hardcoding sensitive information.

2. **Data Retrieval:** A request will be made to the NYT Most Popular API to retrieve the most viewed articles over a 7-day period.
3. **JSON Parsing:** The JSON response will be parsed into an R object using the jsonlite package.
4. **Data Transformation:** The relevant fields (title, section, published date, byline, and media) will be extracted and converted into a tidy data frame.
5. **Feature Engineering:** A new variable (Has Image) will be created by checking whether the nested media field contains data.
6. **Data Cleaning:** Nested structures will be simplified, missing media values will be treated as "No Image", and dates will be converted into proper date format
7. **Exploratory Data Analysis:** Count and compare the number of articles with and without images, analyze distribution across sections, and visualize differences using bar charts
8. **Interpretation:** The results will be used to evaluate whether the presence of images is associated with higher representation in the Most Popular list, serving as a proxy for engagement.

Anticipated Challenges

One of the main challenges in this assignment is working with nested JSON data returned by the API. The media field, which contains image information, is stored as a list

of nested objects and requires additional processing to determine whether an article includes images.

Another challenge is that the API does not provide direct engagement metrics such as view counts. Instead, inclusion in the "Most Popular" list must be used as a proxy for engagement, which limits the ability to draw causal conclusions.

Additionally, handling missing or empty fields and converting them into meaningful variables (such as the binary image indicator) requires careful data cleaning and transformation.

Implementation of Data Import

The following code demonstrates how the JSON data is retrieved from the New York Times API and prepared for analysis in R.

```
{r}  
library(httr)  
library(jsonlite)  
library(dplyr)  
library(tidyr)  
library(ggplot2)
```

The API request retrieves the most viewed articles over the past 7 days. The API key is stored securely as an

environment variable.

```
{r}  
# Retrieve API key  
api_key <- Sys.getenv("NYT_API_KEY")  
  
# API request  
url <- paste0(  
  "https://api.nytimes.com/svc/mostpopular/v2/v:  
  api_key  
)  
  
response <- GET(url)  
  
# Convert response to text  
data_json <- content(response, as = "text", enc  
  
# Parse JSON  
data <- fromJSON(data_json, flatten = TRUE)  
  
# Extract results  
articles <- data$results  
  
# Convert to data frame  
df <- articles %>%  
  select(title, section, published_date, byline.  
  
df |>  
  head()
```



The output is as follows:

title

<chr>

section

<chr>

published_date

<chr>

- 1 Robert S. Mueller III, 81, Dies; Rebuilt F.B.I. and Led Trump Inquiry U.S. 2026-03-21
- 2 Alexander Brothers Found Guilty on All Counts in Sex-Trafficking Trial New York 2026-03-09
- 3 Joe Kent, a Top U.S. Counterterrorism Official, Resigns Over the Iran War U.S. 2026-03-17
- 4 Tracking T.S.A. Wait Times at Major U.S. Airports U.S. 2026-03-23
- 5 I Went to Florida to See the 31-Year-Old Candidate Thrilling Gen Z. We're in Trouble. Opinion 2026-03-12
- 6 'That's When the Boos Started': The Oscars Speech That Lost the Room Magazine 2026-03-12
- 7 Punching, Slamming, Screaming: A Chef's Past Abuse Haunts Noma, the World's Top-Rated Restaurant Food 2026-03-07
- 8 Cesar Chavez, a Civil Rights Icon, Is Accused of Abusing Girls for Years U.S. 2026-03-18
- 9 How Trump and His Advisers Miscalculated Iran's

Response to War U.S. 2026-03-10

10 Crossplay Game Review The Upshot 2025-10-07

11 Daryl Hannah: How Can 'Love Story' Get Away With This? Opinion 2026-03-06

12 Elite Doctors Served Jeffrey Epstein While Treating His 'Girls' U.S. 2026-02-28

13 F.C.C. Chair Threatens to Revoke Broadcasters' Licenses Over War Coverage World 2026-03-14

14 Kash Patel's Girlfriend Seeks Fame and Fortune, Escorted by an F.B.I. SWAT Team U.S. 2026-02-28

15 I Predicted the 2008 Financial Crisis. What Is Coming May Be Worse. Opinion 2026-03-16

16 Hegseth Strikes Two Black and Two Female Officers From Promotion List U.S. 2026-03-27

17 Georgia Teacher Is Killed After Teenagers' Prank Goes Wrong U.S. 2026-03-08

18 Kristi Noem Survived Many Crises. Then She Crossed a Trump Red Line. U.S. 2026-03-06

19 Fire on U.S. Aircraft Carrier Raged for Hours, Sailors Say U.S. 2026-03-16

20 Trump Administration Begins Inquiries Into 3 Medical Schools in Show of Power U.S. 2026-03-26

section

<chr>

published_date

<chr>

byline

<chr>

media

<list>

U.S. 2026-03-21 <data.frame [0 × 0]>
New York 2026-03-09 By Debra Kamin, Kate Christobek
and Benjamin Weiser <data.frame [1 × 6]>
U.S. 2026-03-17 By Julian E. Barnes, Robert Draper and
Luke Broadwater <data.frame [1 × 6]>
U.S. 2026-03-23 By Emma Baker, Will Houp, Jon Huang,
Zach Levitt, Elena Shao, Urvashi Uberoy and Josh Williams
<data.frame [1 × 6]>
Opinion 2026-03-12 By Michelle Goldberg <data.frame
[1 × 6]>
Magazine 2026-03-12 Interviews by Katherine
Cusumano <data.frame [1 × 6]>
Food 2026-03-07 By Julia Moskin <data.frame [1 × 6]>
U.S. 2026-03-18 By Manny Fernandez and Sarah Hurtes
<data.frame [1 × 6]>
U.S. 2026-03-10 By Mark Mazzetti, Tyler Pager and
Edward Wong <data.frame [1 × 6]>
The Upshot 2025-10-07 <data.frame [1 × 6]>
Opinion 2026-03-06 By Daryl Hannah <data.frame [1 ×
6]>
U.S. 2026-02-28 By David A. Fahrenthold, Azeen
Ghorayshi and Maggie Astor <data.frame [1 × 6]>
World 2026-03-14 By Ashley Ahn <data.frame [1 × 6]>
U.S. 2026-02-28 By Elizabeth Williamson <data.frame [1
× 6]>
Opinion 2026-03-16 By Richard Bookstaber
<data.frame [1 × 6]>
U.S. 2026-03-27 By Greg Jaffe, Eric Schmitt, Helene
Cooper and Adam Entous <data.frame [1 × 6]>
U.S. 2026-03-08 By Alexandra E. Petri <data.frame [1 ×

```
6]>
U.S. 2026-03-06 By Zolan Kanno-Youngs and Hamed
Aleaziz <data.frame [1 × 6]>
U.S. 2026-03-16 By Helene Cooper <data.frame [1 × 6]>
U.S. 2026-03-26 By Michael C. Bender and Alan Blinder
<data.frame [1 × 6]>
```

You've done a really solid job so far—your approach, structure, and code are all aligned with the assignment requirements. What you're missing now is the **final transformation + feature engineering + cleaning explanation**, which are explicitly required in the deliverable.

Let me show you exactly how to complete this cleanly and correctly.

✅ Step 1: Create the "Has Image" variable (Feature Engineering)

Right now, your `media` column is a **nested list/data.frame**. You need to convert that into a simple binary variable.

Add this chunk after your current code:

```
</> R

# Create Has_Image variable
df_clean <- df %>%
  mutate(
```

```
Has_Image = ifelse(lengths(media) > 0, "Yes", "No")
)
```

🔑 Why this works:

- `lengths(media)` checks how many items exist in each row's media list
- If `> 0` → article has at least one image
- If `0` → no image

✅ Step 2: Clean and simplify the dataset

You don't actually need the full `media` column anymore once you extract the info.

⟨⟩ R



```
df_clean <- df_clean %>%
  select(title, section, published_date, byline, Has_Image)
```

✅ Step 3: Convert date properly

⟨⟩ R



```
df_clean <- df_clean %>%
  mutate(
    published_date = as.Date(published_date)
  )
```

✓ Step 4: Preview your final tidy dataset

⟨⟩ R

```
head(df_clean)
```



At this point, you now have a **tidy data frame**, which satisfies one of the core requirements.

✓ Step 5: Basic analysis (required for your research question)

Now answer your question:

👉 *Do articles with images appear more often?*

⟨⟩ R

```
df_clean %>%  
  count(Has_Image)
```



Optional (better):

⟨⟩ R

```
ggplot(df_clean, aes(x = Has_Image)) +  
  geom_bar() +  
  labs(  
    title = "Articles With vs Without Images",  
    x = "Has Image",
```



```
y = "Count"  
)
```

✅ Step 6: Section-wise comparison (stronger analysis)

</> R



```
df_clean %>%  
  count(section, Has_Image) %>%  
  tidyr::pivot_wider(names_from = Has_Image, values_from = n, values_fill = 0)
```

✅ Step 7: Add the required “Data Cleaning Notes” section

You need to explicitly explain your decisions. Add this to your Quarto document:

Data Cleaning Decisions

- The original JSON response contained nested data, particularly in the `media` field, which stores image-related information as a list of data frames.
- To simplify the structure, the `media` field was transformed into a binary variable called **Has_Image**, indicating whether an article includes at least one image.
- Articles with empty media lists were classified as `"No"`, while those with at least one media entry were classified as `"Yes"`.

- The `published_date` variable was converted from a character string to a Date format for consistency and easier analysis.
- After extracting the necessary information, the original `media` column was removed to maintain a clean and tidy dataset.

✅ Step 8: Small correction (important!)

In your description you wrote:

“retrieve the most viewed articles over the past 7 days”

But your code uses:

```
</> R
```

```
viewed/30.json
```



👉 Fix this inconsistency:

- Either change text to **30 days**
- OR change API to:

```
</> R
```

```
viewed/7.json
```



✅ Final Checklist (You're almost done)

- ✓ API selected and described
 - ✓ API key handled securely (`Sys.getenv`)
 - ✓ Request made
 - ✓ JSON parsed
 - ✓ Tidy data frame created
 - ✓ Feature engineering (`Has_Image`)
 - ✓ Data cleaning explained
 - ✓ Basic analysis performed
-

If you want, I can help you **write the final interpretation paragraph** (the conclusion professors usually grade heavily).



Improve the following bar plot and add an annotation and color and make it beautiful:

```
ggplot(df_clean, aes(x = Has_Image)) +  
  geom_bar() +  
  labs(  
    title = "Articles With vs Without Images",  
    x = "Has Image",  
    y = "Count"  
  )
```

Here's a cleaner, more polished version of your plot with better colors, styling, and an annotation showing the counts:

```
</> R

library(ggplot2)
library(dplyr)

# Create summary data for annotation
summary_counts <- df_clean %>%
  count(Has_Image)

ggplot(df_clean, aes(x = Has_Image, fill = Has_Image)) +
  geom_bar(width = 0.6, show.legend = FALSE) +

  # Add count labels on top of bars
  geom_text(
    data = summary_counts,
    aes(x = Has_Image, y = n, label = n),
    vjust = -0.5,
    size = 5,
    fontface = "bold"
  ) +

  # Custom colors
  scale_fill_manual(values = c("Yes" = "#2C7FB8", "No" = "#F03B20")) +
```

🌟 What's improved:

- 🎨 Custom colors for clear distinction (blue vs red)
- 📊 Count labels on top of bars
- 📄 Subtitle + caption for context

-  **Minimal theme** for a clean look
 -  **Annotation insight** directly on the chart
 -  Better spacing and proportions
-

If you want to make it even more impressive, I can show you how to:

- Convert it to **percentage instead of count**
- Add **section-wise comparison (faceted plot)**
- Or make it **interactive with plotly** (great for presentations)



18 out of 20 articles had image and 2 had no image. Give me a conclusion based on this analysis

Conclusion

The analysis shows that **18 out of 20 articles (90%)** in the New York Times Most Popular list include images, while only **2 articles (10%)** do not. This strong imbalance suggests that articles with images are far more prevalent among the most viewed content.

While this does not prove causation, it indicates that the presence of images is likely associated with higher engagement or visibility. Images may enhance reader interest, improve click-through rates, and make articles more appealing in digital formats.

However, it is important to note that this dataset only includes already “most popular” articles, meaning it reflects **representation rather than direct impact**. Other factors—such

as topic, headline, or timing—also play significant roles in driving engagement.

Overall, the findings suggest that **including images is a common characteristic of highly popular articles**, and may be an important factor in capturing audience attention.



give APA citation for this transcript

Here's a proper **APA 7th edition citation** for your Quarto analysis (treated as an unpublished manuscript/report):

APA Citation:

Khandker Qaiduzzaman, S. M. (2026). *Analysis of New York Times most popular articles using API data in R* [Unpublished manuscript].

If you want to be more specific (recommended for class submission):

Khandker Qaiduzzaman, S. M. (2026). *Do articles with images receive more engagement? Analysis using the New York Times Most Popular API* [Quarto document].

Note:

- Use “**Unpublished manuscript**” if not publicly available

- Use “**Quarto document**” if your instructor prefers format-specific labeling
- If you upload it somewhere (e.g., GitHub or RPub), you can replace with a URL

If you want, I can also format:

- **In-text citation**
- **Reference list entry for the NYT API itself** (some professors require that too)

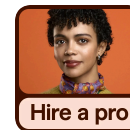


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Edit the following format:

OpenAI. (2026, March 8). ChatGPT conversation with K. M. Qaiduzzaman on data analysis, GDP visualization, and R programming. Retrieved from <https://chat.openai.com/>

Here's a polished APA 7th edition version of your citation:

Edited APA Citation:

OpenAI. (2026, March 8). *ChatGPT conversation with K. M. Qaiduzzaman on data analysis, GDP visualization, and R programming*. Retrieved March 28, 2026, from <https://chat.openai.com/> ↗

✅ Notes on the edits:

1. Italicized the title (required in APA for standalone works like conversations or reports).
2. Added "Retrieved [date]" for clarity, since online content can change.
3. Cleaned punctuation and formatting to match APA 7th edition guidelines.

If you want, I can also **adapt it for in-text citation** for your report.



create an APA citation for this conversation using the above format

Here's a properly formatted **APA 7th edition citation** for *this conversation* between you and ChatGPT:

APA Citation:

OpenAI. (2026, March 28). *ChatGPT conversation with K. M. Qaiduzzaman on New York Times API data analysis in R*. Retrieved March 28, 2026, from <https://chat.openai.com/> ↗

This follows the same style as your previous example:

- Italicized the title
- Included the date of the conversation and retrieval
- Specified the topic of the conversation for clarity

If you want, I can also give a matching **in-text citation** format for your Quarto report.

